Achilles Repair Protocol Phase 1

Purpose

Maximum Protection

Typical Time Frame

0-2 Weeks

Precautions

- Avoid passive DF and any stretching on the graft
- CAM boot with wedge (3 wedges)
- NWB

Criteria to Pass

- N/A for this phase - often patient has not even come into the clinic at this point

Priority List & Suggested Exercises

1. **Edema Control**
    a. Elevated/Ice
2. **Protect the Graft and Ensure Closure of Incision**
3. **Ankle ROM (boot off)**
    a. Gentle ankle circles
    b. Ankle pumps
    c. Foot intrinsics: short foot, toe spreading, toe curling, toe lifts, towel scrunch
4. **Core & Lower Extremity Strength (all with boot on)**
    a. 4 way SLR
    b. Clamshells
    c. Side plank holds
    d. Leg extension
    e. HS curls

Special Considerations

- BFR: can use for LE strengthening to maintain while NWB
- Can perform seated upper body activities to avoid loss of strength
- Consider aerobic conditioning for HEP throughout all phases to maintain aerobic capacity with UE UBE

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Achilles Repair Protocol Phase 2

Purpose

Protection

Typical Time Frame

2-8 Weeks

Precautions

- WBAT with crutches, CAM boot, and wedge (1 wedge removed every 1-2 weeks beginning at 2 weeks)
- AROM/PROM DF to neutral ONLY
- No scar tissue work until incision is healed

Criteria to Pass

- Pain < 5/10
- Incision closed
- DF neutral passively
- Full ankle ROM (PF, Ev, Inv) within 95% of uninvolved ankle
- Ambulate in CAM boot without crutches

Priority List & Suggested Exercises

1. **Ankle ROM**
   a. Ankle pumps> exaggerated ankle circles> alphabets
   b. Seated heel slides (to DF neutral)
   c. Joint mobilizations: improve accessory motions at talocrural, subtalar, distal tibiofibular, midfoot, and forefoot joints as needed
   d. Great toe ROM
2. **Foot/Ankle Strength**
   a. Low level ankle isometrics holds start with 10 sec work up to 30 sec (all directions limiting DF to neutral) > band resisted ankle Ev, Inv
   b. Seated Baps Board (no weight and start with smaller ball and progress to bigger)
   c. Foot intrinsic strengthening exercises short foot, toe spreading, toe curling, toe lifts, towel scrunch
   d. Seated Calf Raise Progression
3. **Lower Extremity Strength**
   a. Continue phase 1 exercises while progressing load
   b. Double leg Shuttle Press
4. **Gait Training**
   a. Initiate WB with crutches in the boot > d/c crutches
   b. Weight Shifting (side to side) > out of boot

Special Considerations

- Swelling: compression, elevation, ice, retrograde massage
- BFR: continue to use for LE strengthening while NWB
- Incision: Should be closed by 6 weeks
- Programming: High volume for exercises 2-3 sets of 12-15 reps for open chain LE exercises
- Cardio: Bike with boot

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Achilles Repair Protocol Phase 3

Purpose

Functional Restoration

Typical Time Frame

6-14 Weeks

Precautions

- No aggressive DF stretching
- No IASTM over scar
- Begin phase in CAM boot > shoe with wedge > shoe without wedges (removing a wedge each week)

Criteria to Pass

- AROM DF to 5° in long sitting
- Knee to wall test within 4cm of the uninvolved side
- 5/5 ankle strength (DF, Inv, Ev)
- 10 double leg calf raises (straight knee)
- SLS x 30sec (straight and soft bent knee (20-30°))
- Y balance reach 80% of uninvolved side without hip strategy
- Normal gait pattern ambulating without the boot or wedge

Priority List & Suggested Exercises

1. **Ankle ROM**
   a. Ankle circles > ankle pumps > alphabets
   b. DF: Seated heel slides, gentle gastroc & soleus stretch w/ stretch rope, ½ kneeling knee to wall to first resistance
   c. Joint mobilizations: talocrural, subtalar, distal tibiofibular, midfoot, and forefoot
   d. Other: Great toe ROM, Tibial rotation
2. **Foot/Ankle Strength**
   a. Band resisted ankle PF, DF, Ev, Inv isometric holds
   b. Resisted isotonic PF, DF, Ev, Inv, diagonal patterns
   c. Seated Baps Board > Standing Baps Board
   d. Double leg calf raise holds (UE assistance if necessary)
   e. Shuttle Press Calf Raise Progression
3. **Balance Training**
   a. Tandem stance
   b. SLS on wedge > SLS on floor
   c. Star drill on wedge
4. **Functional Strength**
   a. TRX assisted squats on wedge with symmetrical form > TRX squats without heels elevated > air squat
   b. Double Leg Wall sit progression on Wedge
   c. Modified Step Up Progression
   d. LE strengthening: split squat w/ wedge, lateral split squat with wedge, hamstring ball curls, shuttle press, prone hs curls
5. **Gait Training**
   a. Weight shifting
   b. TM walking (no incline)
   c. Retro walking, side to side

Special Considerations

- BFR: continue to use for LE strengthening to maintain while NWB
- Pain: Remain <4/10 during PT and 24 hours following
- Programming:
  - High volume for exercises 2-3 sets of 12-15 reps for open chain exercises
  - For all calf strengthening exercises volume should be 2-3 sets of 8 > 12 reps
  - Continue full body exercise for patients as time allows and as appropriate for patient
- Once they are cleared to walk with a regular shoe and no wedge, emphasize performing exercises barefoot for majority of the session
- Train B LE including the calf muscles throughout all phases of rehab

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Achilles Repair Protocol Phase 4

Purpose

Progressive Overload

Typical Time Frame

12-20 Weeks

Precautions

- Initiate passive stretching of the calf/achilles gently and for short durations
- No sled pushes or training in positions in maximum DF.

Criteria to Pass

- Knee to wall test within 2cm of uninvolved side
- Y balance within 95% of uninvolved side
- 10 SL calf raises (80% of available ROM)
- 30 lateral step downs (8" step, back flat against wall) with no hip compensation
- 15 forward step downs (8" step)
- 25 SL squats to box (20") with minimal trunk flexion
- 25 SL bridges on bench (knee bent at 20°)

Priority List & Suggested Exercises

1. **Ankle ROM**
   a. DF: Closed chained DF ankle circles, Banded DF on step, ½ kneeling knee to wall to first resistance, Standing gastroc & soleus stretch (to 1st resistance)
   b. Other: Tibial rotations
2. **Foot/Ankle Strength**
   a. Standing Baps Board (big ball)
   b. Standing Calf Raise Progression
   c. Loaded seated single leg calf raises
3. **Balance Training**
   a. SLS on unstable surface > SLS KB pass > SLS w/ MB diagonals
   b. Star drill > Star drill on unstable surface
   c. SL RDL reaches > SL RDL reaches with MB
   d. Tandem walking
4. **Lower Extremity Strength**
   a. Air squat (symmetrical to 90°) > Goblet squat
   b. Reverse slider lunge (TRX assisted) > no assistance > weighted
   c. Step up progression
   d. LSD progression
   e. Fwd Step Down progression
   f. SL squat progression
   g. Split squats w/ wedge
   h. Lateral split squats with wedge
   i. Shuttle Press Variations
   j. Sled Pulls
   k. Hamstring ball curls
   l. Prone hamstring curls
   m. DB RDLs
   n. Bridge Progression

Special Considerations

- BFR: Continue use in the earlier part of PT session
- Scar Mobilization: Begin using IASTM directly to the tendon around 16 weeks
- Programming:
  - Continue UE and LE exercises with OKC machines; Train in the hypertrophy phase to limit atrophy and deconditioning
  - For calf raise strengthening exercises work up to 3x15 for double leg and 3x8 for SL before progressing via weight or depth
- Prioritize performing exercises barefoot for partial to majority of the session
- Conditioning: Perform cardio exercises including bike, elliptical, retro walking

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Achilles Repair Protocol Phase 5

Purpose

Intro to Impact

Typical Time Frame

16-24 Weeks

Precautions

- Avoid overstretching of the Achilles
- Don’t initiate running until patient has passed the criteria below for return to jogging

Criteria to Pass

To Begin Jogging: (Use Strengthlevel.com)

- Able to perform 25 SL calf raises through 90% of available ROM
- 12 reps at 70% Novice Goblet Squats
- 12 reps at 70% Novice Forward Lunges
- 12 reps at 70% Novice DB RDLs

To Move Onto Phase 6:

- Able to perform and tolerate run walk program for 2 weeks without significant pain or swelling

Priority List & Suggested Exercises

1. **Foot/Ankle Strength**
   a. Standing Baps Board > add weight
   b. Single leg calf raise hold variations
   c. Heel walks/Toe walks > increase distance > add weight > add distance
   d. Standing calf raise progression > Deficit calf raise progression
   e. Loaded seated calf raises
   f. Toe raises (Back against wall) > Tib bar
2. **Agility/Impact**
   a. Agility ladder (basic coordination)
   b. Mini hops > Jogging progression > hop & stop > UE assisted treadmill intervals
   c. Deceleration: snap down progression > cable decels
   d. Jump progression
3. **Lower Extremity Strength**
   a. Squat variations
   b. Lunge progression
   c. Deadlift/RDL variations
   d. Lateral squat progression
   e. Farmer Walks
   f. Loaded step ups and step downs

Special Considerations

- BFR: Continue use in the earlier part of PT session
- Scar Mobilization: Begin using IASTM/Cupping directly to the tendon around 16 weeks
- Programming:
  - Continue UE and LE exercises with OKC machines; Train in the hypertrophy phase to limit atrophy and deconditioning
  - For calf raise strengthening exercises work up to 3x15 for double leg and 3x8 for SL before progressing via weight or depth
- Consider using a 10° incline wedge under heels when you introduce “load” with functional strengthening exercises then wean off
- Patient moves to maintenance routine for ROM unless they are still struggling w/ DF

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Achilles Repair Protocol Phase 6

Purpose

Deceleration

Typical Time Frame

4-6 Months

Precautions

- Don’t initiate cutting until patient has passed the criteria below for return to cutting

Criteria to Pass

To Begin Cutting: (Use Strengthlevel.com)

- 8 reps at 80% of Intermediate Goblet Squats
- 8 reps at 80% of Intermediate Forward Lunges
- 8 reps at 80% of Intermediate DB RDLs
- Decel efficiently in the sagittal plane
- 15 SL hops in place w/ good load acceptance & landing in same spot

To Move Onto Phase 7:

- Sprint w/o compensation
- ATHLETES: Run 1mi w/o limping
- GENERAL POP: Jog 5min w/o limping
- Land symmetrical <10% w/ 5 squat jumps on Force Plates

Priority List & Suggested Exercises

1. **Agility/Impact**
   a. Decel Progression: multiplanar hops > ladder decels > change of pace > stop & goes
   b. Medicine ball drops
   c. 3-way hurdle hop
   d. Hop over hurdle in sagittal plane > hurdle hop into lateral plane
   e. Hop over hurdle (lateral plane) > transition hops
   f. Lateral shuffles, reactive cutting, pro agility drill, T-test drill
   g. Cutting footwork drill > cone cutting > reactive cone cutting
2. **Foot/Ankle Strength**
   a. Toe raises (Back against wall) > Tib bar
   b. Toe walks > increase distance > add weight > add distance
   c. Single leg calf raise hold variations
   d. Deficit Calf Raise Progression
   e. Calf Raise Variations
3. **Lower Extremity Strength**
   a. Continue with Phase 5 exercises
   b. Add more load to functional movements including step ups, step downs, squats, split squats, hinges, etc.
   c. Exxentric squat, split squat, Bulgarian split squats

Special Considerations

- Icing at the end of session or at home/Continue DF stretching
- Programming:
  - At least 3 days off
  - No > 3-4 sets of 3-5 reps agility
  - Plyometrics should be done prior to strengthening in order to avoid fatigue
  - Once patient can tolerate advanced plyometrics for multiple sets/reps without pain and with good form you can begin supersetting strength with power
  - Tempo: 0-0-0 Decelerate the load and transition to acceleration as fast as possible
- Conditioning through cross training: elliptical, bike, rower, jogging

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Achilles Repair Protocol Phase 7

Purpose	Precautions
Return to Sport	• Monitor swelling and pain after exercises; running, and plyometrics
Typical Time Frame	
6 Months +	

Criteria to Pass

- SL hop test >90% LSI with average of 3 trials
- Triple hop test >90% LSI with average of 3 trials
- <10% deficit for Deceleration RFD and Peak Force with landing w/ SL Countermovement Jump on Force Plates
- SL drop vertical jump RSI > .5
- Girth measurements of the calf within 95% of uninvolved limb
- Score of 48 or > on the Injury-Psychological Readiness to Return to Sport (I-PRRS) Questionnaire

Priority List & Suggested Exercises

1. **Agility/Impact**
   a. Depth Jump Progression
   b. Hurdle Hop Variations
   c. Lateral shuffles, reactive cutting, pro agility drill, T-test drill
   d. Cutting footwork drill > cone cutting > reactive cone cutting
   e. Sprinting drills
   f. Jump progression
2. **Foot/Ankle Strength:**
   a. Weighted baps board
   b. Weighted toe walks
   c. Calf Raise variations (loaded)
3. **Functional Strength**
   a. Sled pushes
   b. Continue with strengthening exercises from phase 5 and consider Complex training
   c. Complex training: Compound Exercise at 80% 1RM (4 reps) followed by a plyometric jump exercise that mimic the same motor pattern (3 reps); 3-6 sets
4. **Conditioning**
   a. Cross train rather than just running, so you do not overstress Achilles

Special Considerations

- Prioritize Recovery: 2-3 days off/wk
- Programming:
  - For calf raise strengthening exercises begin to focus on hypertrophy parameters 4-6 sets between 6-12 reps with increasing loads
  - Plyometric sets should range from 3-5 sets and reps from 5-1 depending on level of difficulty/load. Higher the load the fewer the reps
- Icing at the end of session or at home/Continue DF stretching

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